

Fundamentals principle of photocatalysis

Barkha Rani^{a,b}, Arpan Kumar Nayak^c, and Niroj Kumar Sahu^b

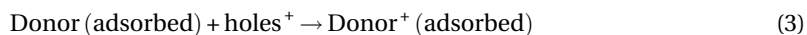
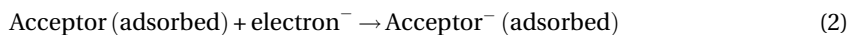
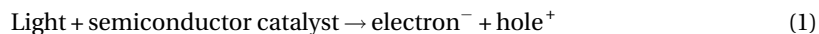
^aSCHOOL OF ELECTRONICS ENGINEERING, VELLORE INSTITUTE OF TECHNOLOGY, VELLORE, TAMIL NADU, INDIA ^bCENTRE FOR NANOTECHNOLOGY RESEARCH, VELLORE INSTITUTE OF TECHNOLOGY, VELLORE, TAMIL NADU, INDIA ^cDEPARTMENT OF PHYSICS, SCHOOL OF ADVANCED SCIENCES, VELLORE INSTITUTE OF TECHNOLOGY, VELLORE, TAMIL NADU, INDIA

1 Introduction

Photocatalysis is a combination of two words: photo and catalysis. Photo stands for light and catalysis is a process to enhance the chemical reaction rate by dropping the activation energy for the primary reaction to begin [1]. In the catalysis process, a substance is introduced in the reaction which enhances the rate of transformation of reactant to a product without being altered in the end [2]. These substances are known as catalysts, which are semiconducting in nature [3]. A fundamental distinction between conventional catalysis and photocatalysis is the usage of photons or light in place of heat for the activation of the catalyst. Hence, photocatalysis is a process of activating the catalyst by the application of light, which further improves the rate of the reaction without being self-involved.

Photocatalytic oxidation or the advanced oxidation process (AOP) has been studied extensively from the past five decades for the elimination of harmful microorganisms, pigments, contaminants, and toxic substances from polluted water [2–7]. In this process, the molecular transformation or photo-driven chemical reaction takes place in the catalyst or its surfaces. The aforementioned process is divided into two categories: sensitized and catalyzed photoreaction depends on whether the first excitation happens at the catalyst, surface or on the adsorbate molecules [8]. A sensitized photoreaction occurs when photoexcitation initiates in the catalyst substrate and then the energy is transferred in the form of an electron (e^-) which migrates from the valence band (VB) to the conduction band (CB). In contrast, a catalyzed photoreaction occurs in an adsorbate molecule which further reacts with a catalyst substrate. The adsorption of reactants and the absorption of photons are the two key factors of the photocatalysis process. The photo sorption and desorption can take place either in a solid catalyst or liquid phase. In the case of the solid phase, irradiation of photon energy equal to or greater than the bandgap energy (E_g) of the catalyst results in the formation of e^- and h^+ pairs. A photoelectron is generated in the CB whereas photogenerated h^+ presents in the VB of the catalyst. This photogenerated

electron-hole pair (PEHP) reacts with oxygen and hydroxyl molecules to form highly active radicals. These radicals further react with toxic reactants and transform them into nontoxic ones [9]. However, in the fluid phase, an instantaneous absorption occurs and e^- and h^+ transfer take place in between acceptor and donor molecules, depending on the redox potential of each adsorbate. The e^- migrates toward acceptor molecules whereas positive h^+ migrates toward donor molecules and formed reactive ions. These reactions are explained below:



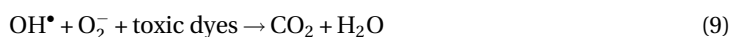
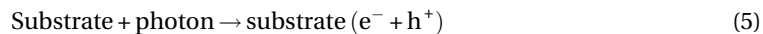
These ions and free radicals are subsequently reacted to form intermediate and final products. Sometimes, the efficiency of the photoreaction is reduced due to the recombination of PEHP. This recombination converts the photoelectrical energy into heat.



In photocatalysis, radiation is used for the acceleration of chemical reactions. The radiation is broadly divided into two regions: ultraviolet and visible, which are prominently selected based on the catalyst E_g . Only 4% of the solar spectrum comes under the UV region. Hence, the fundamental principles, mechanism, kinetics, and parameters that adversely impact on the photoactivity of the catalytic process along with the strategy to utilize solar spectrum for photocatalysis are explained in detail in this chapter.

2 Basic principle for photocatalytic reactions

In a photocatalytic reaction, a photocatalyst acts as a substrate that gets activated by the application of photon light and subsequently produces highly active radicals. The generated radicals and ions react further with toxic organic molecules and break them into CO_2 and H_2O . Generally, transition oxide-based materials like ZnO , TiO_2 , SnO_2 , etc. are used as the substrate, and water and oxygen are used as the adsorbate [10]. The reactions involved in the process of dye removal are as follows:



The abovementioned equations suggest the necessity of interaction between the photocatalyst and adsorbed (acceptor or donor) molecules. Sometimes these interactions

are induced by either electrostatic forces or chemical bonding such as dipole-dipole interaction, van der Waal forces, ion exchange, hydrogen bonding, and adsorbed hydrophobicity. It can be observed from Eq. (5) that the irradiation of photon energy ($h\nu \geq E_g$) causes the creation of e^- and h^+ pairs in the photoactive materials. The energy to generate a photoexcited pair in a catalyst is directly related to the wavelength (λ) of the light by this relation:

$$\lambda \leq \frac{hc}{E_g} \quad (10)$$

$$\lambda \leq \frac{1240}{E_g} \quad (11)$$

where c and h are the velocity of light and Plank's constant, respectively. The wavelength in Eq. (11) is in nanometer units. The overall photocatalytic reaction is based on two mechanisms: direct or indirect photocatalytic mechanisms. These mechanisms are explained in detail below.

2.1 Direct photocatalytic principle

Two different models (the Langmuir-Hinshelwood and Eley-Rideal processes) have been used to describe the direct photocatalytic pathway of the reaction. In the Langmuir-Hinshelwood (L-H) model, the photocatalyst is activated by photons and produces e^- in CB and h^+ in VB. Furthermore, the adsorbed species act as traps for photogenerated h^+ and create a reactive radicals state on the surface of the catalyst, which further decays due to the recombination of e^- and h^+ . This recombination again regenerates the catalyst material. The abovementioned model provides the relation between the reaction rate and initial solute concentration [11, 12]. The L-H model can be explained by the following relation [13]:

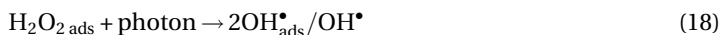
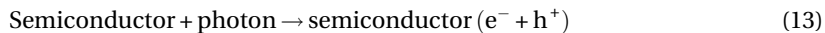
$$1/r = 1/K_r + 1/K_r K_a C \quad (12)$$

where r is the reaction rate (mg/L min), K_r is the specific reaction rate constant (mg/L min), K_a is the equilibrium constant (l/mg), and C is the concentration of dye. However, in the Eley-Rideal (E-R) model, the free carriers get trapped by surface defects. The degradation path followed in E-L process is as follows: (i) trapping of photogenerated h^+ by surface defect (s); (ii) formation of surface-active centers (s^+); (iii) reaction between s^+ and chemisorbed dye ($\text{dye}_{\text{ads}}^+$); (iv) formation of adduct species such as $(s\text{-dye}_{\text{ads}})^+$; (v) decomposition of $(s\text{-dye}_{\text{ads}})^+$; and (vi) production of product and surface defect sites [12].

2.2 Indirect photocatalytic principle

In this method, the PEHP is generated on the surface of the semiconductor. The photo-induced h^+ reacts with the adsorbed water molecules or moisture to form hydroxyl

radicals. Moreover, the PEHP is allowed to produce H_2O_2 , which further breaks and produces more reactive radicals ($\text{OH}^\bullet$) for the oxidation of dye molecules [14]. The overall steps can be explained as follows:



3 Reaction pathways for photogenerated electron-hole pair

3.1 Pathways of PEHP within a semiconducting catalyst

The irradiation of photon energy over catalyst surface results in a number of reactions given in Fig. 1A. Initially, the semiconducting photocatalyst absorbs the photon energy which results in the excitation across the bandgap depending upon whether $h\nu \geq E_g$. The e^- transfers to CB and a positive h^+ is generated in VB. PEHP has a sufficient time in the range of nanoseconds to undergo charge transfer to adsorbed species on the catalyst surface from solution or gas phase contact [15]. During de-excitation, the e^- and h^+ can follow various paths such as: (i) migration of e^- or h^+ to the photocatalyst surface for the transfer of e^- to adsorbed organic or inorganic species or solvent; (ii) surface or volume recombination of PEHP which releases heat; (iii) donation of e^- at the catalyst surface to reduce O_2 to ions; and (iv) migration of h^+ to the surface where an e^- from a donor species can combine with the surface h^+ , thus oxidizing the donor species. These migrations and transformations depend upon the e^- and h^+ position and redox potential level of the adsorbate species [8]. The potential level of acceptor species should be lower than the CB potential of the semiconductor. However, the band position of donor species should be above the VB to donate an e^- to the vacant h^+ . The band edge position of the photo-active material changes with the changing pH of the electrolyte with respect to the redox potential of the adsorbate molecules [8].

3.2 Charge carrier trapping

The reaction with the adsorbed species and efficient charge transfer is the mandatory condition for PEHP. The lifetime of PEHP before recombination falls in the range of microseconds to nanoseconds. It can be increased by trapping the charge carriers, which delays the

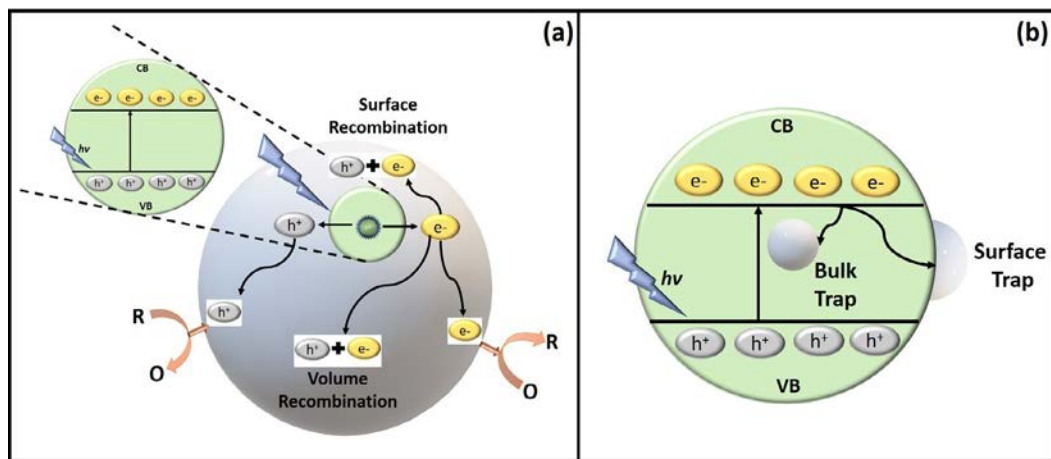


FIG. 1 Schematic of (A) photoexcitation and de-excitation in a catalyst material and (B) surface/bulk charge carrier trapping. Concept adapted from A. L. Linsebigler, G. Lu, J. T. Yates, *Photocatalysis on TiO₂ surfaces: principles, mechanisms, and selected results*, *Chem Rev* 95 (1995) 735–758.

recombination process by a fraction of a nanosecond. Synthesized colloidal or polycrystalline photocatalysts exhibit some defects in their crystal structure or crystal lattice; they also have some irregularity in their surface and bulk. These distortions result in the formation of e^- states at the surface, which are distant compared to the bulk semiconductor's energy band. The abovementioned states work as a trap for PEHP and reduce their recombination. There are mainly two types of traps present in the semiconducting catalyst: bulk traps and surface traps (Fig. 1B) [8]. The trap sites for both models are localized and present within the E_g of the catalyst. The PEHP become trapped in particular sites and form localized states either in bulk or at the surface. The population of traps depends upon the energy difference between the CB and traps. Trapping of photoexcited e^- minimizes the entropy of the system [16].

4 Fundamental principles based on the type of catalyst

Photocatalysts exhibit properties such as: (i) efficient absorption of photons; (ii) effective charge separation of PEHP; (iii) facile migration or diffusion of mobile charges; and (iv) rapid charge transfer, which results in the fast transformation of chemical species. These characteristics depend upon the material and vary greatly with changes to the material specification. For example, the light absorption process of the semiconductor catalyst is different than that of the plasmonic catalyst. Moreover, the lifetime of PEHP in semiconductors is in the range of picoseconds to microseconds whereas it is more than 100 femtoseconds for plasmonic materials [17]. Therefore, a detailed study of the principle that governs the photocatalytic process in different classes of material such as bulk semiconductors, quantum dots, and two-dimensional material is crucial.

4.1 Semiconductor-based photocatalyst

The photocatalytic principle for semiconductor material has been widely researched and studied extensively. In n-type semiconductors, the concentration of e^- is higher than that of h^+ . The addition of an n-type semiconductor in the electrolyte results in a more negative value of Fermi level (E_F) than the electrochemical potential of the electrolytes. This causes the formation of the Schottky junction [18]. Under dark conditions, e^- starts migrating from the semiconductor toward the electrolyte, which creates the equilibrium at the interface. This results in a bending of the band upward at the electrode/electrolyte interface and looks like the Schottky junction, which forms at the semiconductor/metal interface. Illumination results in the photoactivation of the catalyst surface and generation of PEHP. This PEHP is separated in the depletion region due to the presence of an internal built-in field (E_b). Doping level and degree of band bending are the deciding factors for the width (W) of the depletion region [19]. Here, the charges are removed to either the bulk or the solid/liquid interface with the help of E_b [20]. The diffusion length (D) of the mobile charge is the distance from the depletion region and it depends solely on the lifetime and mobility of the charge carriers [21]. The total number of charges produced within the D and the depletion region can only participate in the chemical reactions. This collective width ($W + D$) is associated with the penetration depth of the photons and subsequently provides the adsorption coefficient (α) of the material. High efficiency is only observed in those materials which have a higher value of total width ($W + D$) than optical depth ($1/\alpha$). A similar phenomenon can be correlated to p-type materials by only spinning the polarities of mobile charge carriers [20].

4.2 Quantum dot-based photocatalyst

Energy state quantization is a prominent phenomenon observed in the quantum dot-based photocatalyst. The reason for this is the size of the material which is comparable with the Bohr radius of the material. It gives perfect alignment of the energy state of the charge carriers to the molecular orbital of the targeted reactants. This quantization provides an additional advantage in terms of selective oxidation [22, 23]. Zhao et al. [22] reported the selective oxidation of alcohols for hydrogen production by CdSe quantum dots (QDs). Moreover, these QDs are used as sensitizers and frequently used as cocatalysts where they act as light absorbers for the second catalyst material. The quantization of electronic states results in the tuning of optical adsorption edges. Generally, the photoactivation of QDs results in the generation of e^- which is injected into the CB of the supporting semiconductor. However, photogenerated h^+ is present in the VB of the QD and requires a sacrificial e^- donor group. This case creates the internal built-in barrier at the interface of the QD/semiconductor, which significantly increases the lifetime of charge carriers [24]. An increase in an E_g (1.9–2.4 eV) is observed in CdSe with the change in size. The change in the size of CdSe makes CB more negative whereas VB is constant throughout the variation. This causes a substantial potential difference between the QD and semiconducting TiO_2 surface, and increases the charge transfer rate of the reaction [24].

4.3 Two-dimensional material-based photocatalyst

Separation and migration of the charge carriers in the two-dimensional material are different from the bulk semiconductor and quantum dot-based photocatalyst. The reduction in the thickness and the layered morphology results in a change in the bandgap as well as the presence of anisotropic properties in the two-dimensional materials. Li et al. [25] demonstrated the different charge separation in the layered BiOCl particles. Different crystal orientation causes a variation in charge separation and further changes the rate of reaction as well as the lifetime of the charge carriers [26, 27]. Moreover, defects present in the two-dimensional material affect the charge distribution and subsequently a drastic change is seen in the charge behaviors. For example, an oxygen vacancy is created in the two directions (010) and (001) of the BiOCl, causing a change in the surface potential. Furthermore, water molecules get adsorbed at the vacancy sites of the BiOCl-001 whereas dissociation adsorption appears on the vacancy sites of the BiOCl-010 surface. Xie et al. reported a similar approach for improving the charge separation in ZnIn_2S_4 [28]. Zhang et al. [29] described the photoactivity of a 2D composite in which $\text{Zn}_{0.8}\text{Cd}_{0.2}\text{S}$ was deposited over the reduced graphene oxide. The composite showed an easier migration of e^- from the CB of $\text{Zn}_{0.8}\text{Cd}_{0.2}\text{S}$ to rGO layers.

5 Factors affecting the photocatalysis reaction

The efficiency, rate, and percentage degradation of photocatalytic reaction depend upon several factors such as the quantity of catalyst, dye concentration, pH of the solution, temperature, additives, wavelength, and intensity of the light source. The effects of these factors are explained in detail.

5.1 Effect of quantity of catalyst

The mass of photocatalyst directly influences the initial rate of the reaction. A greater number of catalysts results in the presence of a huge number of adsorption sites, which further increases the rate of the photocatalytic reaction [30, 31]. However, after a high dose of catalyst, the reaction rate becomes independent of the mass. This range relies on the working condition and geometry of the photoreactor [32]. A higher amount of catalyst results in the masking of photosensitive surfaces, which hinders the efficient absorption of photons as well as reducing the light penetration in the reactor [33–35]. Furthermore, agglomeration of the particles is also observed in the case of high doses which lowers the effective surface area of the catalyst [36]. Herrmann [32] performed the reaction in two different reactors (batch reactor and CPC solar reactor) and optimized the amount of TiO_2 catalyst for the degradation process. He found that a small quantity of TiO_2 (0.2 g/L) is enough to give higher efficiency in a CPC solar reactor than the batch reactor (2.5 g/L). He also concluded that the optimum amount of photocatalyst depends upon the initial concentration of solute. Rauf et al. [37] varied the amount of $\text{TiO}_2/\text{V}_2\text{O}_5$

catalyst from 10 to 40 mg/20 mL and found that the percentage of decolorization is increased with increasing dose and again starts falling. This increment is attributed to the presence of a large number of active sites for dye adsorption. In addition, it causes the generation of a higher number of hydroxyl radicals for the reaction. Saquib et al. [38] and Qamar et al. [39] also reported that the degradation rate increases with increasing catalyst dosage.

5.2 Effect of wavelength and intensity of light

The wavelength and intensity of light sources also have an impact on the photocatalytic reaction. Photoexcitation energy equal to or more than the E_g is required for the transition of e^- from VB to CB. TiO_2 has an E_g of 3.2 eV, thus it requires a wavelength of a minimum of 400 nm (UV) for photoactivation of the catalyst. Ollis et al. [40] divided the intensity of a light source into three regions (low, intermediate, and high) and explained their effect on the catalytic process. Ollis stated that at low intensity (0–20 mW/cm^2), the rate of reaction is increased linearly with increasing light intensity and follows the first order of the reaction. At intermediate state, it follows the half order reaction, i.e., the rate of the reaction relies on the square root of the light intensity. However, an independency in the rate of reaction is observed at high intensity due to the complete utilization of active sites of the material. Neppolian et al. [41] performed the degradation of three textile dyes using the TiO_2 catalyst by irradiating solar and UV light. They showed that UV light is more adequate than solar irradiation. They also varied the intensity of solar radiation and found that the percentage degradation increases with increasing light intensity. High intensity of light results in the formation of more e^- - h^+ along with a negligible recombination rate. At lower intensities, PEHP separation competes with recombination, causing the slow formation of free radicals which results in less percentage degradation of the dyes [42]. Rani et al. [43] used SnO_2 nanoparticles as a catalyst for the removal of methylene blue dye. They examined the effect of wavelength by irradiating 254 and 365 nm light and found that the efficiency is higher in the case of 254 nm wavelength. This behavior was attributed to the presence of a wider absorption band which further results in the generation of a greater number of PEHP. Liu et al. [44] performed the experiment in three different intensities and showed that the decolorization of azo dye increases with increasing light intensity. Increasing light intensity also enhances the decolorization rate [45, 46].

5.3 Effect of pH

The photocatalytic process is governed by three reactions: (i) direct oxidation by the positive charge carrier; (ii) direct reduction by negative charge carrier; and (iii) hydroxyl radical attack. All these reactions depend upon the nature of the substrate and pH of the solution [47]. The pH of the solution tunes the electric double layer and changes the surface charges at the solid-liquid interface. This modification further changes the potential of catalytic reaction, absorption-desorption process, and the separation of PEHP at the

surface of the catalyst. Metal oxide and hydroxide materials exhibit an amphoteric nature which develops either positive or negative charge at the surface [48–51]. Hence, variation in the pH of the solution shows an impact on the adsorption process [49]. Dye degradation is restrained by the H^+ ions in the acidic medium, which subsequently reduces the efficiency of the photocatalytic process. However, in a basic medium, a greater number of hydroxyl ions are present which nullifies the acidic by-product generated during the dye degradation process as well as potentially resulting in the formation of more hydroxyl radicals. The nature of dye is equally crucial for its percentage degradation at higher pH. Some of the dyes are highly stable in the basic medium. Their chromophores are intact even after the irradiation of light and consequently the degradation rate decreases [52]. Bubacz et al. [53] examined methylene blue degradation and found that the rate of degradation increases with an increase in pH. Rani et al. [43] observed that the SnO_2 nanoparticles show fast degradation in the basic medium. This behavior is due to the electrostatic interaction between the anionic catalyst and cationic dye, which results in the high adsorption of the dye molecules over the catalyst surface [54, 55]. Baran et al. [56] reported a sixfold increment in adsorption efficacy during the acidification of the solution from pH 8 to 4.5. This increment is attributed to a change in charge of the dye (bromocresol purple). Many studies reported the high efficiency of positive TiO_2 catalyst in an acidic medium compared to an alkaline medium [57, 58]. They observed more adsorption of dye molecules at lower pH due to the presence of a negative charge in their structure [59].

5.4 Effect of the initial concentration of dye

Generally, the efficiency of dye degradation reduces with increasing concentration of dye [33]. The high concentration of dye results in more adsorption of dye over the catalyst surface. This hinders photons in reaching the catalyst surface and thus reduces the generation of hydroxyl radicals for the photocatalytic reaction. Rauf et al. observed that the percentage decolorization of dye decreases with increasing dye concentration [37]. Neppolian et al. [41] worked on three different dyes (Reactive Yellow 17, Reactive Red 2, and Reactive Blue 4) and studied the effect of dye concentration on the catalytic rate. They found that percentage degradation decreases with increasing concentration in all cases. This result is in agreement with other studies [60–63]. Various reasons reported by the literature for less activity are: (i) absorption of a photon by dye molecules hinders the penetration of light to catalyst surface [59, 62, 64]; (ii) blocking of active sites by dye molecules results in the inefficient generation of hydroxyl radicals [65]; (iii) dye molecules work as a barrier between hydroxyl radicals, PEHP, and catalyst, which further reduces the reaction PEHP to adsorbed species [48]; (iv) high concentration shields the photon and resultantly the path length of the photons reduces; and (v) lesser generation of hydroxyl radicals causes less availability of free radicals for reacting dye molecules.

The reaction of the hydroxyl radicals with dye molecules directly influences the rate of the reaction. Initially, the rate of dye degradation increases with increasing the initial concentration of dye due to the presence of more dye molecules and it follows the first-order kinetics [66]. In this case, dyes are being adsorbed on the surface of the catalyst and forming

monolayers which do not restrict the penetration of photons. However, the rate is maximal and shows zero-order kinetics after reaching a critical level of dye concentration [32, 59]. A similar pattern is found by Saquib et al. [38] and Kiriakidou et al. [59]. Saquib et al. experimented by varying the dye concentration from 0.18 to 0.5 mM and they observed that the rate increased with increasing concentration up to 0.25 mM and then abtained.

5.5 Effect of temperature

Temperature also has a significant impact on the rate of photocatalytic reaction. Recombination of charge carriers enhances with an improvement in temperature. Moreover, the desorption process of adsorbed species is increasing with the temperature which results in less catalytic activity of the materials. The photocatalytic reactor does not require any heating element and works at room temperature. The true activation energy (E_T) and apparent activation energy (E_A) are two factors that are used for explaining the effect of temperature on the catalytic reaction. At moderate temperatures (20–80°C), E_T is nil and E_A is very small. However, an increment in E_A is observed at low temperatures (–40°C to 0°C), which is further used to heat the adsorption species [32]. The heating of the solution causes an increase in the rate constant of the photodegradation process. However, lowering the temperature results in a spontaneous exothermic reaction which favors the adsorption of reactant species. It also supports the adsorption of the final product. On the other side, when the temperature reaches the boiling point of the water, the reaction becomes limited and does not support the adsorption of reactant species [67]. Soares et al. [68] found that the optimum temperature for the photocatalytic reaction is 40–50°C for a TiO_2 catalyst.

5.6 Effect of additives

Efficiency and photocatalytic rate are also influenced by the presence of additives in the solution matrix. Additives are initially added to the dye solution for improving the industrial process. They are ionic compounds and present as an ion in the dye solution such as Zn^{2+} , Na^+ , Fe^{2+} , Cl^- , persulfate ions, and so on. However, the abovementioned addition had a negative impact on the efficiency of the photocatalytic reaction. For example, Fe^{2+} ions reacted with highly active hydroxyl radicals and formed OH^- and Fe^{3+} [69]. This results in lowering the concentration of hydroxyl radicals and thus decreases the efficiency of the photocatalytic process. Numerous studies have reported the scavenging effect of h^+ and $\text{OH}^\bullet$ due to the presence of Cl^- , SO_4^{2-} , CO_3^{2-} , and HCO_3^- [37, 70, 71]. Rauf et al. [37] reported the e^- scavenging by bromate ions which further reduces the degradation efficiency of the reaction. They showed that the bromate ions reacted with six e^- and formed bromide ions which further react with hydroxyl radicals and minimize the percentage degradation of dye molecules.

However, some oxidants such as H_2O_2 and O_3 increase the rate of reaction by trapping the e^- , which further prevents the recombination of PEHP. Moreover, they contribute to the generation of more traps e^- as well as oxidizing radicals for the degradation process.

Even air and oxygen have significant impacts on the dye degradation process [72–74]. Photogenerated e^- reacts with oxygen and forms superoxide ions, which further react with adsorbed species and complete the degradation process. However, in the absence of oxygen the percentage degradation decreases, and the recombination rate increases. Oxygen gets adsorbed on the surface of the catalyst and acts as a trap for photogenerated e^- . On the other side, hydrogen peroxide increases the generation of $OH^\bullet$ either by self-decomposition or reduction at CB [46]. The low concentration of H_2O_2 favors the efficient formation of $OH^\bullet$ and restrains the recombination of PEHP. However, at high concentration, it reacts with the excess H_2O_2 present in the solution.

5.7 Quantum size effects

The reduction in the size of semiconductor catalyst (<10 Angstrom) shows the quantization effect. These effects are prominent when the size of the particle is comparable to the De Broglie wavelength of the charge carriers. The PEHP induced in these particles experiences confinement in potential as well as in geometry. This confined PEHP does not experience bulk electronic states (CB and VB); instead, it produces a quantization of discrete electronic states and increases the effective E_g of the catalyst [75, 76]. Many researchers have examined the size-dependent properties of semiconducting catalysts and compared them with the bulk properties [76–78]. Brus theoretically and experimentally explained how effective E_g is changing with size [78]. Reduction in the size of CdS from 100 to 26 Angstrom results in a blue shift in absorption and emission spectra, which indicates the increase in effective E_g of the materials (Fig. 2). Wang et al. [79] showed that the reduction in the size of PbS from bulk to 13 Å results in an increase in E_g by six factors from 0.4 eV (bulk).

5.8 Effect of other parameters on the photoactivity of catalyst

Surface area, particle size, lattice mismatch, distorted structure, and presence of impurity are the critical factors that influence the catalytic activity of the semiconducting material. These factors have a significant impact on the adsorption process as well as on the recombination rate of PEHP [80]. The large surface area of the catalyst results in the presence of more active sites for the adsorption of dye molecules and thus promotes the speed of the reaction [81]. Muggli et al. [82] claim that the TiO_2 particles with mixed (rutile and anatase) phases are more efficient than the individual one.

6 Strategies for enhancement of photocatalytic properties of semiconducting materials

The efficiency of the photocatalysis process is determined by the quantum yield, i.e., the number of reactions involved per photon absorbed. These reactions mainly depend on: (i) the E_g of photocatalyst; (ii) the rate of generation of PEHP; and (iii) the fate of migration.

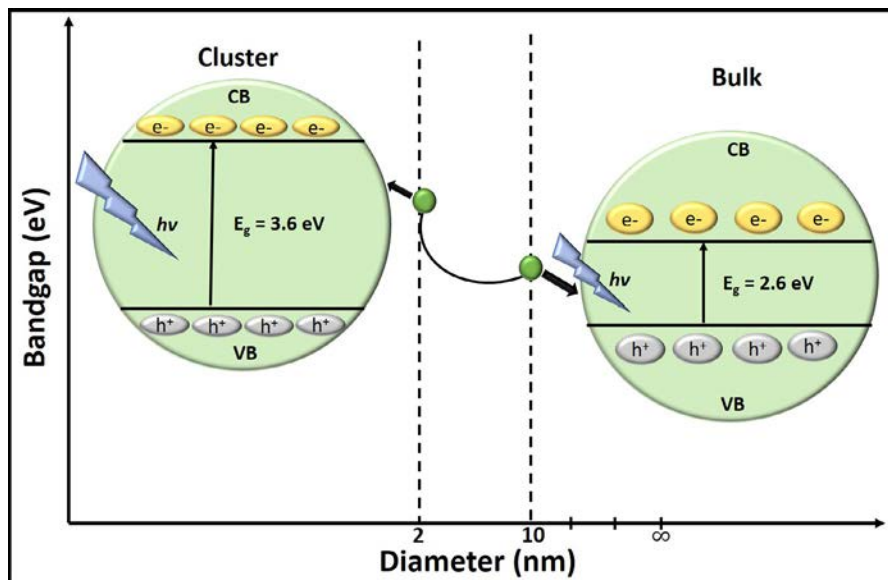


FIG. 2 Effect of quantum size on CdS bandgap. Concept adapted from A. L. Linsebigler, G. Lu, J. T. Yates, *Photocatalysis on TiO₂ surfaces: principles, mechanisms, and selected results*, *Chem Rev* 95 (1995) 735–758.

The rate of generation of PEHP depends on the irradiation condition (photon energy, flux) and optical absorption characteristics (surface area, E_g , size), whereas the fate of migration depends on whether the reaction occurs at the surface of the particles or within the photocatalyst. Surface recombination of PEHP results in the generation of heat, which reduces the efficiency of the photocatalytic process. This recombination can be decreased by increasing the crystallinity of the lattice, reducing catalyst size, or increasing the transfer rate [8]. If the material is highly crystalline, then there are minimal chances of the presence of an impurity, surface defects, and bulk defects, whereas the reduction in the size induces the point defects in the particles which further act as traps for e^- or h^+ and reduce the rate of recombination. These trapped present in the localized energy level in the E_g region and trapped the charge carriers for a microsecond and nanosecond [83]. The efficiency of the reaction can be improved further by preadsorption of the species on the photocatalyst surface [84]. Moreover, the addition of dopants, metals, or forming a heterojunction also reduces the recombination of PEHP, thereby increasing the efficiency of the photocatalytic process.

Until now, TiO₂-based photocatalysts have been extensively used for the removal of dye or toxic containments from wastewater [85]. TiO₂ has a wide E_g of 3.2 eV and photoactivation is only possible by the irradiation of UV light, which employs only 4% of the solar spectrum [86]. Currently, researchers are more focused on the usage of visible light-driven photocatalyst (VLDP) since it covers the largest fraction of the solar spectrum. Bi₂O₃ [87], WO₃ [88], Bi₂WO₆ [89], CdS [90], and C₃N₄ [91] are a few examples of VLDP. Although many experiments have been performed on the preparation of VLDP, their applications are still

limited due to the constrained absorption of visible light and short PEHP lifetime. Various strategies have been adapted to overcome these issues, such as doping, textural design, or constructing a heterojunction by coupling with secondary substance. Among these, the formation of a heterojunction is more efficient for improving the efficiency of photocatalytic reactions [92–94]. The formation of a heterojunction catalyst can be categorized into four parts: (a) a semiconductor-semiconductor heterojunction (Sc-Sc H) which creates sufficient built-in potential (V_b) for an oxidation-reduction reaction as well as providing high absorption of the solar spectrum; (b) a semiconductor-metal heterojunction (Sc-M H) which causes an effective transition and separation of PEHP; (c) a semiconductor-carbon group heterojunction (Sc-C H) which results in sufficient migration and separation of PEHP; and (d) a multicomponent heterojunction (McH) which improves the absorption capacity of material and also broaden the range for UV/visible light absorption. The principles of all these categories are illustrated in Figs. 3 and 4.

6.1 Semiconductor-semiconductor heterojunction (Sc-Sc H)

p-n Sc-Sc H and non-p-n Sc-Sc H are two class of Sc-Sc H. Construction of a p-n junction provides advantages in terms of: (i) high adsorption capacity; (ii) efficient collection and separation of charge carriers; (iii) longer lifetime of the PEHP; and (iv) fast charge transfer to the catalyst. Moreover, it separates the localized inappropriate oxidation and reduction reactions in the nano space. Basically, when n-type and p-type semiconductors come in contact, they create a space charge region at the interface due to the diffusion of e^- and h^+ . This results in the generation of V_b which further directs the e^- and h^+ to migrate in the

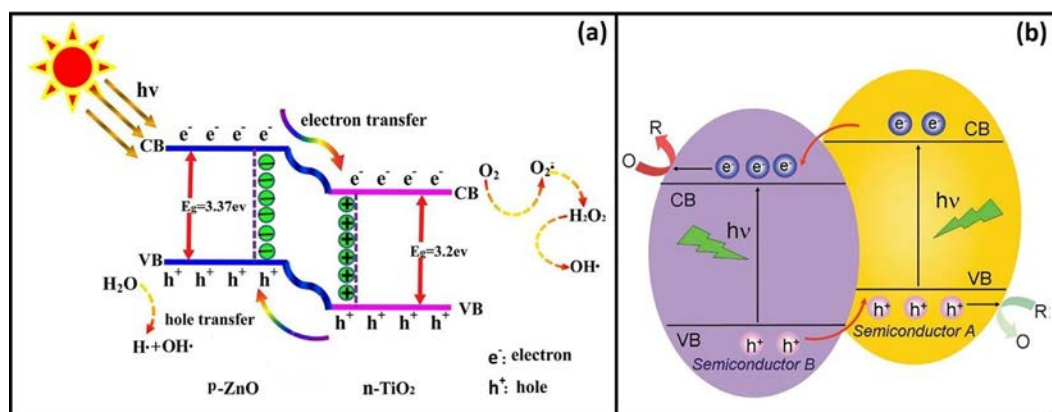


FIG. 3 Schematic structure of (A) p-n semiconductor-semiconductor heterojunction and (B) non-p-n semiconductor-semiconductor heterojunction. Reprinted with permission from H. Wang, L. Zhang, Z. Chen, J. Hu, S. Li, Z. Wang, et al. *Semiconductor heterojunction photocatalysts: Design, construction, and photocatalytic performances*. *Chemical Society Reviews* 2014;43:5234–5244. <https://doi.org/10.1039/c4cs00126e> and R. Zha, R. Nadimicherla, X. Guo, *Ultraviolet photocatalytic degradation of methyl orange by nanostructured TiO₂/ZnO heterojunctions*, *J Mater Chem A* 3 (2015) 6565–6574

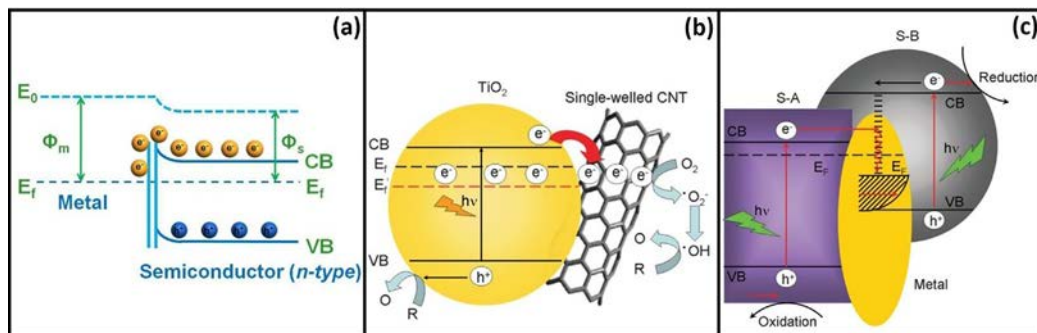


FIG. 4 Schematic structure of (A) semiconductor-metal heterojunction, (B) semiconductor-carbon heterojunction, and (C) multicomponent heterojunction. Reprinted with permission from H. Wang, L. Zhang, Z. Chen, J. Hu, S. Li, Z. Wang, J. Liu, X. Wang, *Semiconductor heterojunction photocatalysts: design, construction, and photocatalytic performances*, *Chem Soc Rev* 43 (2014) 5234–5244 and S. Liu, C. Han, Z. Tang, Y. Xu. *Heterostructured semiconductor nanowire arrays for artificial photosynthesis*. *Materials Horizons* 2016;3:270–282. <https://doi.org/10.1039/c6mh00063k>.

opposite direction. The irradiation of light in the p-n heterojunction with the energy higher than the E_g of the catalyst causes a quick separation of PEHP by the built-in electric field. This electric field directs the electrons to migrate to the CB of the n-type semiconductor and h^+ to the VB of the p-type semiconductor [95]. Another type of heterojunction formed in the semiconductors is non-p-n-type, in which both semiconductors (A and B) forming the heterojunction are the same types, i.e., either n-type or p-type. Here, two different semiconductors with matched potential are bonded together to form the heterostructures. If the CB level of A is higher than the CB of B, then the transition of e^- from the CB of A to the CB of B is observed in the heterostructure with irradiation of photons. However, when the VB of A is higher than the VB of B, the h^+ starts migrating from the B to the A semiconductor. This results in the presence of a vast number of e^- on the surface of the B semiconductor and h^+ on the A semiconductor. Hence, efficient separation of PEHP by the internal field is obtained in the photocatalyst, which further reduces the recombination rate and promotes the photocatalysis process. Shang et al. [96] reported enhanced photocatalytic activity in Bi_2WO_6 - TiO_2 by eight orders compared to bare Bi_2WO_6 under visible light irradiation. Wang et al. [97] demonstrated higher efficiency in SnO_2 - TiO_2 heterostructure toward Rhodamine B degradation than bare SnO_2 and TiO_2 . Wang et al. [98] showed the enlargement in the absorption range due to the formation of a Bi_2O_3 - Bi_2WO_6 heterojunction which results in a 2.7-fold increase in the degradation rate of Bi_2WO_6 .

6.2 Semiconductor-metal heterojunction (Sc-M H)

Another way to increase charge separation and reduce recombination of PEHP is by forming an Sc-M H. In this type of contact, e^- start to flow from one catalyst to another depending on their Fermi levels. Generally, the work function of the metal is higher than the n-type semiconductors such as TiO_2 and SnO_2 , and e^- start to migrate to the metal from

semiconductor to balance the Fermi energy levels. This creates the Schottky barriers at the interface where the semiconductor has excess h^+ charge whereas the metal has excess e^- . Moreover, this barrier acts as a trap for e^- and reduces the recombination of PEH. Wang et al. [99] reported high decomposition of methylene orange dye by Ag-AgCl. They compared their result with N-doped TiO_2 and found eight times higher activity for dye degradation. Lu et al. [100] compared their finding with bare AgBr and N-doped TiO_2 . They found that the activity of Ag-AgBr increased 1.5 times to AgBr and 13 times to N-doped TiO_2 for the removal of Rhodamine B dye under visible light irradiation. Improved activity in these materials is mainly due to: (i) the localized surface plasmons resonance effect; (ii) better quantum efficiency; and (iii) improved light absorption.

6.3 Semiconductor-carbon heterojunction (Sc-C H)

Various types of carbon material such as graphene, reduced graphene oxide, activated carbon, and CNT have been used for making heterojunctions. Carbon-based materials exhibit high surface area, which results in the adsorption of more pollutants and thus improves the photoactivity of the catalyst material [101]. Graphene and CNT have a metallic structure and formed a Schottky barrier at the interface of semiconductor/CNT. This junction causes an improvement in the recombination time. Apart from this, CNT allows more photoexcited e^- from a semiconductor due to its huge e^- storage capacity which hinders the recombination of PEHP [3]. Generally, photoexcitation of the catalyst results in the generation of e^- in the semiconductor. Electrons presents in the space-charge regions start migrating from the CB of semiconductor to CNTs for balancing the Fermi level, whereas h^+ present in the semiconductor continuously participates in the redox reaction [102]. The formation of a heterojunction with graphene provides key advantages to semiconductor photocatalyst such as: (i) high conductivity; (ii) high surface area; (iii) extended absorption wavelength; (iv) promoted charge transfer; and (v) restrained recombination of PEHP. These advantages result in higher photocatalytic activity in the composite material compared to the bare one. Moreover, the interaction of graphene with semiconductors causes a shift in the Fermi level, which further reduces the CB potential of the semiconducting material. This reduction allows the fast migration of photoinduced e^- and suppresses the recombination of charge carriers. Gao et al. [103] reported the Bi_2WO_6 incorporated graphene sheet as a photocatalyst for the removal of Rhodamine B dye. They found that the composite exhibited three times higher activity than the bare Bi_2WO_6 under visible light irradiation. This behavior was mainly attributed to the efficient absorption of photons, fast e^- transportation, and the reduction in the CB potential. A similar result was also reported by Du et al. [101]. They reported an 11-fold increase in the activity of TiO_2 -graphene compared to the bare TiO_2 for removal of methylene blue dye.

6.4 Multicomponent heterojunction (McH)

Various types of materials are extensively studied and developed for efficient use of the solar spectrum. But they are suffering in terms of a limited area for visible light photoresponse. In

this direction, the McH-based system plays a significant role in which two or more VLDPs and an e^- migrating system are integrated [104, 105]. Generally, two semiconductors have different absorption ranges and can be excited by visible or UV light. The combination of these two semiconductors results in the widening of photoexcitation of the material. The irradiation of photons in the catalyst causes the generation of e^- and h^+ in both semiconductors, and two types of e^- transfer are observed in these systems. Firstly, the photoexcited e^- of semiconductor A start migrating to the metal through Schottky barriers because the CB of semiconductor A is higher than the Fermi level of metal. The rate of transferring e^- in semiconductor A is faster than the recombination of PEHP. Consequently, a massive number of h^+ are present in the VB of the semiconductor for participating in the redox reaction, and the life span of PEHP also increases [106]. On the other side, the Fermi level of metal is above the VB of semiconductor B, which results in the migration of h^+ from semiconductor B to the metal and reduces the recombination of PEHP in semiconductor B. This results in the presence of a greater number of photoexcited e^- in the CB of semiconductor B which further react with O_2 and air, and generate O_2^- . Hence, the use of visible/UV light results in both the generation of PEHP in the semiconductors and the simultaneous e^- transfer from semiconductor A to the metal to semiconductor B. Here, conjugation of the metal in the semiconductors provides several advantages in terms of: (i) storage system; (ii) recombination center for h^+ in the VB of semiconductor B and e^- in the CB of semiconductor A; (iii) improving the interfacial charge transfer; and (iv) a complete separation of e^- in CB of semiconductor B and h^+ in the VB of semiconductor A which further participated in the redox reaction. Tada et al. [104] demonstrated dye degradation by using the multicomponent system CdS-Au-TiO₂, in which CdS and TiO₂ act as semiconductors A and B, and Au is conjugated with them. Tada et al. reported higher activity in the McH than the single component (CdS or TiO₂) or binary components (Au-TiO₂, CdS-TiO₂).

7 Conclusion

In this chapter, the basic principle for photocatalytic reaction, pathways followed by photoinduced charge carriers, and various strategies to improve the photocatalytic activity of semiconductors were explained in detail. The fundamental principle and mechanisms based on the class of catalysts such as semiconductors, quantum dots, and two-dimensional structures were also discussed. Moreover, the impacts of various parameters such as pH, initial concentration of dye, amount of catalyst, and the intensity of wavelength on the rate of reaction were explained. The formation of heterostructure increases the percentage degradation of dye as well as providing efficient usage of the solar spectrum by broadening the absorption range of the photocatalyst and enhancing the separation and life span of charge carriers. This overview should help the reader to develop a stable VLDP with a high degree of oxidation for the removal of toxic dyes.

References

- [1] Qu Y, Duan X. Progress, challenge and perspective of heterogeneous photocatalysts. *Chem Soc Rev* 2013;42:2568–80.
- [2] Legrini O, Oliveros E, Braun AM. Photochemical processes for water treatment. *Chem Rev* 1993;93:671–98.
- [3] Hoffmann MR, Martin ST, Choi W, Bahnemann DW. Environmental applications of semiconductor photocatalysis. *Chem Rev* 1995;95:69–96.
- [4] Carey JH, Lawrence J, Tosine HM. Photodechlorination of PCB's in the presence of titanium dioxide in aqueous suspensions. *Bull Environ Contam Toxicol* 1976;16:697–701.
- [5] Rani B, Thamizharasan G, Nayak AK, Sahu NK. Degradation mechanism of organic dyes by effective transition metal oxide. In: *Photocatalysts in Advanced Oxidation Processes for Wastewater Treatment*. Wiley; 2020. p. 197–228.
- [6] Reiche H, Bard AJ. Heterogeneous photosynthetic production of amino acids from methane-ammonia-water at platinum/titanium dioxide. Implications in chemical evolution. *J Am Chem Soc* 1979;101:3127–8.
- [7] Fujishima A, Honda K. Electrochemical photolysis of water at a semiconductor electrode. *Nature* 1972;238:37–8.
- [8] Linsebigler AL, Lu G, Yates JT. Photocatalysis on TiO₂ surfaces: principles, mechanisms, and selected results. *Chem Rev* 1995;95:735–58.
- [9] Low J, Yu J, Jaroniec M, Wageh S, Al-Ghamdi AA. Heterojunction Photocatalysts. *Adv Mater* 2017;29:1601694.
- [10] Thongam DD, Gupta J, Sahu NK. Effect of induced defects on the properties of ZnO nanocrystals: surfactant role and spectroscopic analysis. *SN Appl Sci* 2019;1:1030.
- [11] Silva CG, Faria JL. Photochemical and photocatalytic degradation of an azo dye in aqueous solution by UV irradiation. *J Photochem Photobiol A Chem* 2003;155:133–43.
- [12] Serpone N, Emeline AV. Suggested terms and definitions in photocatalysis and radiocatalysis. *Int J Photoenergy* 2002;4:943970.
- [13] Machado AEH, Miranda JA, Freitas RF, Duarte ETFM, Ferreira LF, Albuquerque YDT, Ruggiero R, Sattler C, Oliveira L. Destruction of the organic matter present in effluent from a cellulose and paper industry using photocatalysis. *J Photochem Photobiol A Chem* 2003;1:231–41.
- [14] Yue B, Zhou Y, Xu J, Wu Z, Zhang X, Zou Y, Jin S. Photocatalytic degradation of aqueous 4-chlorophenol by silica-immobilized polyoxometalates. *Environ Sci Technol* 2002;36:1325–9.
- [15] Nosaka Y, Fox MA. Kinetics for electron transfer from laser-pulse irradiated colloidal semiconductors to adsorbed methylviologen: dependence of the quantum yield on incident pulse width. *J Phys Chem* 1988;92:1893–7.
- [16] Rothenberger G, Moser J, Graetzel M, Serpone N, Sharma DK. Charge carrier trapping and recombination dynamics in small semiconductor particles. *J Am Chem Soc* 1985;107:8054–9.
- [17] Fan W, Zhang Q, Wang Y. Semiconductor-based nanocomposites for photocatalytic H₂ production and CO₂ conversion. *Phys Chem Chem Phys* 2013;15:2632–49.
- [18] Gelderman K, Lee L, Donne SW. Flat-band potential of a semiconductor: using the Mott-schottky equation. *J Chem Educ* 2007;84:685.
- [19] Yang X, Wang D. Photophysics and photochemistry at the semiconductor/electrolyte Interface for solar water splitting. *Semicond Semimet* 2017;97:47–80.
- [20] Lin Y, Xu Y, Mayer MT, Simpson ZI, McMahon G, Zhou S, Wang D. Growth of p-type hematite by atomic layer deposition and its utilization for improved solar water splitting. *J Am Chem Soc* 2012;134:5508–11.

- [21] Hodes G, Kamat PV. Understanding the implication of carrier diffusion length in photovoltaic cells. *J Phys Chem Lett* 2015;6:4090–2.
- [22] Zhao L, Meng Q, Fan X, Ye C, Li X, Chen B, Ramamurthy V, Tung C, Wu L. Photocatalysis with quantum dots and visible light: selective and efficient oxidation of alcohols to carbonyl compounds through a radical relay process in water. *Angew Chem Int Ed* 2017;56:3020–4.
- [23] Olshansky JH, Ding TX, Lee YV, Leone SR, Alivisatos AP. Hole transfer from photoexcited quantum dots: the relationship between driving force and rate. *J Am Chem Soc* 2015;137:15567–75.
- [24] Robel I, Kuno M, Kamat PV. Size-dependent Electron injection from excited CdSe quantum dots into TiO₂ nanoparticles. *J Am Chem Soc* 2007;129:4136–7.
- [25] Li H, Li J, Ai Z, Jia F, Zhang L. Oxygen vacancy-mediated Photocatalysis of BiOCl: reactivity, selectivity, and perspectives. *Angew Chem Int Ed* 2018;57:122–38.
- [26] Li R, Zhang F, Wang D, Yang J, Li M, Zhu J, Zhou X, Han H, Li C. Spatial separation of photogenerated electrons and holes among {010} and {110} crystal facets of BiVO₄. *Nat Commun* 2013;4:1432.
- [27] Yu J, Low J, Xiao W, Zhou P, Jaroniec M. Enhanced photocatalytic CO₂-reduction activity of Anatase TiO₂ by coexposed {001} and {101} facets. *J Am Chem Soc* 2014;136:8839–42.
- [28] Yang W, Zhang L, Xie J, Zhang X, Liu Q, Yao T, Wei S, Zhang Q, Xie Y. Enhanced photoexcited carrier separation in oxygen-doped ZnIn₂S₄ nanosheets for hydrogen evolution. *Angew Chem Int Ed* 2016;55:6716–20.
- [29] Zhang J, Yu J, Jaroniec M, Gong JR. Noble metal-free reduced graphene oxide-Zn_xCd_{1-x}S nanocomposite with enhanced solar photocatalytic H₂ - production performance. *Nano Lett* 2012;12:4584–9.
- [30] Herrmann JM. Heterogeneous photocatalysis: an emerging discipline involving multiphase systems. *Catal Today* 1995;24:157–64.
- [31] Muruganandham M, Swaminathan M. Photocatalytic decolourisation and degradation of reactive Orange 4 by TiO₂-UV process. *Dyes Pigments* 2006;68:133–42.
- [32] Herrmann JM. Heterogeneous photocatalysis: fundamentals and applications to the removal of various types of aqueous pollutants. *Catal Today* 1999;53:115–29.
- [33] Macedo LC, Zaia DAM, Moore GJ, Santana HD. Degradation of leather dye on TiO₂: a study of applied experimental parameters on photoelectrocatalysis. *J Photochem Photobiol A Chem* 2007;185:86–93.
- [34] Kamble SP, Sawant SB, Pangarkar VG. Batch and continuous photocatalytic degradation of Benzenesulfonic acid using concentrated solar radiation. *Ind Eng Chem Res* 2003;42:6705–13.
- [35] Gogate PR, Pandit AB. A review of imperative technologies for wastewater treatment I: oxidation technologies at ambient conditions. *Adv Environ Res* 2004;8:501–51.
- [36] Kaneco S, Rahman MA, Suzuki T, Katsumata H, Ohta K. Optimization of solar photocatalytic degradation conditions of bisphenol a in water using titanium dioxide. *J Photochem Photobiol A Chem* 2004;163:419–24.
- [37] Rauf MA, Bukallah SB, Hamadi A, Sulaiman A, Hammadi F. The effect of operational parameters on the photoinduced decoloration of dyes using a hybrid catalyst V₂O₅/TiO₂. *Chem Eng J* 2007;129:167–72.
- [38] Saquib M, Muneer M. TiO₂-mediated photocatalytic degradation of a triphenylmethane dye (gentian violet), in aqueous suspensions. *Dyes Pigments* 2003;56:37–49.
- [39] Qamar M, Saquib M, Muneer M. Photocatalytic degradation of two selected dye derivatives, chromotrope 2B and amido black 10B, in aqueous suspensions of titanium dioxide. *Dyes Pigments* 2005;65:1–9.
- [40] Ollis DF, Pelizzetti E, Serpone N. Photocatalyzed destruction of water contaminants. *Environ Sci Technol* 1991;25:1522–9.

- [41] Neppolian B, Choi HC, Sakthivel S, Arabindoo B, Murugesan V. Solar/UV-induced photocatalytic degradation of three commercial textile dyes. *J Hazard Mater* 2002;89:303–17.
- [42] Bahnemann D. Photocatalytic detoxification of polluted waters. In: Boule P, editor. *Environmental Photochemistry*. Berlin, Heidelberg: Springer Berlin Heidelberg; 1999. p. 285–351.
- [43] Rani B, Punniyakoti S, Sahu NK. Polyol asserted hydrothermal synthesis of SnO₂ nanoparticles for the fast adsorption and photocatalytic degradation of methylene blue cationic dye. *New J Chem* 2018;42:943–54.
- [44] Liu C, Hsieh Y, Lai P, Li C, Kao C. Photodegradation treatment of azo dye wastewater by UV/TiO₂ process. *Dyes Pigments* 2006;68:191–5.
- [45] Sakthivel S, Neppolian B, Shankar MV, Arabindoo B, Palanichamy M, Murugesan V. Solar photocatalytic degradation of azo dye: comparison of photocatalytic efficiency of ZnO and TiO₂. *Sol Energy Mater Sol Cells* 2003;77:65–82.
- [46] So CM, Cheng MY, Yu JC, Wong PK. Degradation of azo dye procion red MX-5B by photocatalytic oxidation. *Chemosphere* 2002;46:905–12.
- [47] Tang WZ, Zhang Z, An H, Quintana MO, Torres DE. TiO₂/UV photodegradation of azo dyes in aqueous solutions. *Environ Technol* 1997;18:1–12.
- [48] Poullos I, Avranas A, Rekliti E, Zouboulis A. Photocatalytic oxidation of Auramine O in the presence of semiconducting oxides. *J Chem Technol Biotechnol* 2000;75:205–12.
- [49] Wang N, Li J, Zhu L, Dong Y, Tang H. Highly photocatalytic activity of metallic hydroxide/titanium dioxide nanoparticles prepared via a modified wet precipitation process. *J Photochem Photobiol A Chem* 2008;198:282–7.
- [50] Guillard C, Lachheb H, Houas A, Ksibi M, Elaloui E, Herrmann J. Influence of chemical structure of dyes, of pH and of inorganic salts on their photocatalytic degradation by TiO₂ comparison of the efficiency of powder and supported TiO₂. *J Photochem Photobiol A Chem* 2003;158:27–36.
- [51] Zielińska B, Grzechulska J, Kalerćzuk RJ, Morawski AW. The pH influence on photocatalytic decomposition of organic dyes over Al₁₁ and P₂₅ titanium dioxide. *Appl Catal B Environ* 2003;45:293–300.
- [52] Tsui SM, Chu W. Quantum yield study of the photodegradation of hydrophobic dyes in the presence of acetone sensitizer. *Chemosphere* 2001;44:17–22.
- [53] Bubacz K, Choina J, Dolat D, Morawski AW. Methylene blue and phenol photocatalytic degradation on nanoparticles of Anatase TiO₂. *Pol J Environ Stud* 2010;19:685–91.
- [54] Ling CM, Mohamed AR, Bhatia S. Performance of photocatalytic reactors using immobilized TiO₂ film for the degradation of phenol and methylene blue dye present in water stream. *Chemosphere* 2004;57:547–54.
- [55] Rani B, Jadhao CV, Sahu NK. Tetramethylene glycol mediated hydrothermal synthesis of defect-rich SnO₂ nanoparticles for fast adsorption and degradation of MB dye. *AIP Conf Proc* 2018;1942:050023.
- [56] Baran W, Makowski A, Wardas W. The effect of UV radiation absorption of cationic and anionic dye solutions on their photocatalytic degradation in the presence TiO₂. *Dyes Pigments* 2008;76:226–30.
- [57] Kansal SK, Kaur N, Singh S. Photocatalytic degradation of two commercial reactive dyes in aqueous phase using Nanophotocatalysts. *Nanoscale Res Lett* 2009;4:709.
- [58] Tanaka K, Padernpole K, Hisanaga T. Photocatalytic degradation of commercial azo dyes. *Water Res* 2000;34:327–33.
- [59] Kiriakidou F, Kondarides DI, Verykios XE. The effect of operational parameters and TiO₂-doping on the photocatalytic degradation of azo-dyes. *Catal Today* 1999;54:119–30.
- [60] Saggiaro EM, Oliveira AS, Pavesi T, Maia CG, Ferreira LFV, Moreira JC. Use of titanium dioxide photocatalysis on the remediation of model textile wastewaters containing azo dyes. *Molecules* 2011;16:10370–86.

- [61] Daneshvar N, Salari D, Khataee AR. Photocatalytic degradation of azo dye acid red 14 in water: investigation of the effect of operational parameters. *J Photochem Photobiol A Chem* 2003;157:111–6.
- [62] Giwa A, Nkeonye PO, Bello KA, Kolawole EG. Solar photocatalytic degradation of reactive yellow 81 and reactive violet 1 in aqueous solution containing semiconductor oxides. *Int J Appl Sci Technol* 2012;2:1–16.
- [63] Zhang T, Oyama TK, Horikoshi S, Hidaka H, Zhao J, Serpone N. Photocatalyzed N-demethylation and degradation of methylene blue in titania dispersions exposed to concentrated sunlight. *Sol Energy Mater Sol Cells* 2002;73:287–303.
- [64] Augugliaro V, Baiocchi C, Prevot AB, Garcí'a-López E, Loddo V, Malato S, Marci' G, Palmisano L, Pazzi M, Pramauro E. Azo-dyes photocatalytic degradation in aqueous suspension of TiO₂ under solar irradiation. *Chemosphere* 2002;49:1223–30.
- [65] Grzechulska J, Morawski AW. Photocatalytic decomposition of azo-dye acid black 1 in water over modified titanium dioxide. *Appl Catal B Environ* 2002;36:45–51.
- [66] Avasarala BK, Tirukkavalluri SR, Bojja S. Synthesis, characterization and photocatalytic activity of alkaline earth metal doped titania. *Indian J Chem Sect A: Inorg, Phys Theor Anal Chem* 2010;49:1189–96.
- [67] Mehrotra K, Yablonsky GS, Ray AK. Macro kinetic studies for photocatalytic degradation of benzoic acid in immobilized systems. *Chemosphere* 2005;60:1427–36.
- [68] Soares ET, Lansarin MA, Moro CC. A study of process variables for the photocatalytic degradation of rhodamine B. *Braz J Chem Eng* 2007;24:29–36.
- [69] Yoon J, Lee Y, Kim S. Investigation of the reaction pathway of OH radicals produced by Fenton oxidation in the conditions of wastewater treatment. *Water Sci Technol* 2001;44:15–21.
- [70] Muruganandham M, Swaminathan M. Photocatalytic decolorization and degradation of reactive orange 4 by TiO₂-UV process. *Dyes Pigments* 2006;68:133–42.
- [71] Rauf MA, Ashraf SS. Fundamental principles and application of heterogeneous photocatalytic degradation of dyes in solution. *Chem Eng J* 2009;151:10–8.
- [72] Alnuaimi MM, Rauf MA, Ashraf SS. A comparative study of neutral red decoloration by photo-Fenton and photocatalytic processes. *Dyes Pigments* 2008;76:332–7.
- [73] Ramirez JH, Maldonado-Hódar FJ, Pérez-Cadenas AF, Moreno-Castilla C, Costa CA, Madeira LM. Azo-dye Orange II degradation by heterogeneous Fenton-like reaction using carbon-Fe catalysts. *Appl Catal B Environ* 2007;75:312–23.
- [74] Dammala P, Machado J, Rani B, Murali S, Devi S, Luwang MN, Sahu NK. Synthesis of biphasic nano-materials based on ZnO and SnO₂: application towards photocatalytic degradation of acid red dye. *Nano-Struct Nano-Objects* 2019;18:100292.
- [75] Kubo R, Kawabata A, Kobayashi S. Electronic properties of small particles. *Annu Rev Mater Sci* 1984;14:49–66.
- [76] Henglein A. Mechanism of reactions on colloidal microelectrodes and size quantization effects. In: Steckhan E, editor. *Electrochemistry II. Topics in Current Chemistry*, 143. Berlin, Heidelberg: Springer; 1988. p. 113–80.
- [77] Spanhel L, Haase M, Weller H, Henglein A. Photochemistry of colloidal semiconductors. 20. Surface modification and stability of strong luminescing CdS particles. *J Am Chem Soc* 1987;109:5649–55.
- [78] Brus LE. Electron–electron and electron-hole interactions in small semiconductor crystallites: the size dependence of the lowest excited electronic state. *J Chem Phys* 1984;80:4403–9.
- [79] Wang Y, Suna A, Mahler W, Kasowski R. PbS in polymers. From molecules to bulk solids. *J Chem Phys* 1987;87:7315–22.
- [80] Thongam DD, Gupta J, Sahu NK, Bahadur D. Investigating the role of different reducing agents, molar ratios, and synthesis medium over the formation of ZnO nanostructures and their photocatalytic activity. *J Mater Sci* 2018;53:1110–22.

- [81] Watson SS, Beydoun D, Scott JA, Amal R. The effect of preparation method on the photoactivity of crystalline titanium dioxide particles. *Chem Eng J* 2003;95:213–20.
- [82] Muggli DS, Ding L. Photocatalytic performance of sulfated TiO₂ and Degussa P-25 TiO₂ during oxidation of organics. *Appl Catal B Environ* 2001;32:181–94.
- [83] Shkrob IA, Sauer MC. Hole scavenging and photo-stimulated recombination of Electron – hole pairs in aqueous TiO₂ nanoparticles. *J Phys Chem B* 2004;108:12497–511.
- [84] Matthews RW. An adsorption water purifier with in situ photocatalytic regeneration. *J Catal* 1988;113:549–55.
- [85] Fujihira M, Satoh Y, Osa T. Heterogeneous photocatalytic oxidation of aromatic compounds on TiO₂. *Nature* 1981;293:206–8.
- [86] Chen C, Cai W, Long M, Zhou B, Wu Y, Wu D, Feng Y. Synthesis of visible-light responsive graphene oxide/TiO₂ composites with p/n heterojunction. *ACS Nano* 2010;4:6425–32.
- [87] Jiang H, Liu J, Cheng K, Sun W, Lin J. Enhanced visible light photocatalysis of Bi₂O₃ upon fluorination. *J Phys Chem C* 2013;117:20029–36.
- [88] Nayak AK, Pradhan D. Microwave-assisted greener synthesis of defect-rich tungsten oxide nanowires with enhanced photocatalytic and photoelectrochemical performance. *J Phys Chem C* 2018;122:3183–93.
- [89] Huang C, Chen L, Li H, Mu Y, Yang Z. Synthesis and application of Bi₂WO₆ for the photocatalytic degradation of two typical fluoroquinolones under visible light irradiation. *RSC Adv* 2019;9:27768–79.
- [90] Hussain W, Malik H, Bahadur A, Hussain RA, Shoaib M, Iqbal S, Hussain H, Green IR, Badshah A, Li H. Synthesis and characterization of CdS photocatalyst with different morphologies: visible light activated dyes degradation study. *Kinet Catal* 2018;59:710–9.
- [91] Wang X, Maeda K, Thomas A, Takanabe K, Xin G, Carlsson JM, Domen K, Antonietti M. A metal-free polymeric photocatalyst for hydrogen production from water under visible light. *Nat Mater* 2009;8:76–80.
- [92] Zhang L, Li J, Chen Z, Tang Y, Yu Y. Preparation of Fenton reagent with H₂O₂ generated by solar light-illuminated nano-Cu₂O/MWNTs composites. *Appl Catal A Gen* 2006;299:292–7.
- [93] Gao H, Wang H, Jin Y, Lv J, Xu G, Wang D, Zhang X, Chen Z, Zheng Z, Wu Y. Controllable fabrication of immobilized ternary CdS/Pt–TiO₂ heteronanostructures toward high-performance visible-light driven photocatalysis. *Phys Chem Chem Phys* 2015;17:17755–61.
- [94] Zhang L, Wong K, Yip H, Hu C, Yu JC, Chan C, Wong P. Effective photocatalytic disinfection of E. coli K-12 using AgBr-ag-Bi₂WO₆ nanojunction system irradiated by visible light: the role of diffusing hydroxyl radicals. *Environ Sci Technol* 2010;44:1392–8.
- [95] Jiang L, Zhou G, Mi J, Wu Z. Fabrication of visible-light-driven one-dimensional anatase TiO₂/Ag heterojunction plasmonic photocatalyst. *Catal Commun* 2012;24:48–51.
- [96] Shang M, Wang W, Zhang L, Sun S, Wang L, Zhou L. 3D Bi₂WO₆/TiO₂ hierarchical Heterostructure: controllable synthesis and enhanced visible photocatalytic degradation performances. *J Phys Chem C* 2009;113:14727–31.
- [97] Wang C, Shao C, Zhang X, Liu Y. SnO₂ nanostructures-TiO₂ nanofibers Heterostructures: controlled fabrication and high photocatalytic properties. *Inorg Chem* 2009;48:7261–8.
- [98] Wang H, Li S, Zhang L, Chen Z, Hu J, Zou R, Xu K, Song G, Zhao H, Yang J, Liu J. Surface decoration of Bi₂WO₆ superstructures with Bi₂O₃ nanoparticles: an efficient method to improve visible-light-driven photocatalytic activity. *CrystEngComm* 2013;15:9011–9.
- [99] Wang P, Huang B, Qin X, Zhang X, Dai Y, Wei J, Whangbo M. Ag@AgCl: a highly efficient and stable photocatalyst active under visible light. *Angew Chem Int Ed* 2008;47:7931–3.

- [100] Lu W, Qin X, Li H, Asiri AM, Al-Youbi AO, Sun X. One-step hydrothermal synthesis of ag nanoparticle decorated submicrometer-scale spherical AgBr colloids: a highly efficient visible light plasmonic photocatalyst for degradation of organic dyes. Part Part Syst Charact 2013;30:67–71.
- [101] Du J, Lai X, Yang N, Zhai J, Kisailus D, Su F, Wang D, Jiang L. Hierarchically ordered macro-mesoporous TiO₂-graphene composite films: improved mass transfer, reduced charge recombination, and their enhanced photocatalytic activities. ACS Nano 2011;5:590–6.
- [102] Woan K, Pyrgiotakis G, Sigmund W. Photocatalytic carbon-nanotube–TiO₂ composites. Adv Mater 2009;21:2233–9.
- [103] Gao E, Wang W, Shanga M, Xua J. Synthesis and enhanced photocatalytic performance of graphene-Bi₂WO₆ composite. Phys Chem Chem Phys 2011;13:2887–93.
- [104] Tada H, Mitsui T, Kiyonaga T, Akita T, Tanaka K. All-solid-state Z-scheme in CdS–au–TiO₂ three-component nanojunction system. Nat Mater 2006;5:782–6.
- [105] Elahifard MR, Rahimnejad S, Haghighi S, Gholami MR. Apatite-coated Ag/AgBr/TiO₂ visible-light photocatalyst for destruction of Bacteria. J Am Chem Soc 2007;129:9552–3.
- [106] Hirakawa T, Kamat PV. Charge separation and catalytic activity of Ag@TiO₂ Core – Shell composite clusters under UV – irradiation. J Am Chem Soc 2005;127:3928–34.